

David Lau

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EDUCATION

SUNY Stony Brook University

Bachelor of Science in Computer Science

Stony Brook, NY

Aug. 2022 – May 2024

Rochester Institute of Technology

Bachelor of Science in Computer Science, transferred

Rochester, NY

Aug. 2020 – May 2022

EXPERIENCE

Software Engineering Intern

BNY Mellon

May 2023 – Aug. 2023

New York, NY

- Developed a financial portal in Angular using an Oracle relational database to aggregate and display earnings of 4 company sectors with charts for enhanced visualization, utilized by over 1,000 internal users.
- Created Maven libraries that reduced code duplication by 40%, improving the readability and scalability of the code base.
- Revamped the web application into an Angular Progressive Web App (PWA), leading to mobile user adoption and improved accessibility across devices as well as decreased load time by 15%.

Research and Outreach Intern

ChangingThePresent

June 2022 – Aug. 2022

New York, NY

- Researched and identified potential strategic partners for the non-profit initiative, resulting in the establishment of partnerships with 12 universities and organization.
- Developed and implemented a comprehensive outreach strategy, resulting in a 25% increase in engagement with potential donors and partners.
- Contributed to a non-profit initiative aimed at channeling part of the \$450 billion spent annually on gifts towards charitable donations.

Tutor

Stem Nation

July 2018 – Feb. 2021

Remote

- Held weekly feedback sessions with students to understand their challenges, leading to more productive tutoring sessions.
- Analyzed students' progress data to identify areas of improvement and tailored instructional strategies to address their specific needs, leading to an improvement in overall student performance.
- Collaborated with a team of tutors to develop and refine curriculum materials, resulting in a decrease in preparation time for subsequent tutoring sessions.

PROJECTS

Geodudes | *React.js, Redux, Leaflet, Node.js, Express, MongoDB*

- Developed GeoDudes, a comprehensive mapping platform, utilizing MERN stack, and redux for efficient state management.
- Integrated an advanced heatmap feature to visualize geospatial data density and also intelligently cluster data points, displaying the count within each cluster for a concise representation of large datasets across the globe.
- Automated deployment to render using GitHub Actions that initiates comprehensive testing using Cypress, Supertest, and Jest, ensuring automatic deployments upon successful push to the main branch.

Zoom Clone | *React.js, Next.js, Tailwind, Clerk, Vercel*

- Developed and deployed a zoom clone, achieving seamless and reliable video communication during meetings.
- Implemented features such as private/public meetings, meeting scheduling, meeting recording, and screen sharing.
- Integrated user authentication with Clerk, supporting both registered users and guest mode, ensuring secure access and personalized user experiences.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, JavaScript, HTML/CSS

Tools & Technologies: Git (GitHub, Gitlab), Jira, Trello, Maven, Spring, MongoDB, SQL, VS Code, IntelliJ

Languages: English, Cantonese